

Shaik Abdus Sattar

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Professional Summary

Computer Science undergraduate specializing in Machine Learning and Deep Learning at the intersection of algorithmic theory, mathematical foundations, image analysis, and efficient programming. Deeply interested in transformer architectures, attention mechanisms, representation learning, and AI systems, with a focus on exploring advanced ML concepts and efficient computational approaches.

Education

Amrita Vishwa Vidyapeetham

Bachelor of Engineering in Computer Science; CGPA: 8.4/10.0

Excellencia Junior College

Intermediate/+2 (State Board): 98%; JEE-Mains: 95.87 percentile

Coimbatore, Tamil Nadu

Aug 2023 – Present

Hyderabad, Telangana

2023

Research Experience

Clinically-Informed Deep Learning for Radiotherapy QA Prediction

2025 – Present

Research Assistant | Advisors: [Dr. Amit Agarwal](#), [Dr. Rose Kamal](#)

- Developing CNN and hybrid deep learning models to predict pre-treatment Gamma passing rates from dose maps and Treatment Plan Complexity Metrics; designing clinically weighted loss functions and structured pipelines for automated radiotherapy QA.

Efficient 3D Molecular Graph Kernels for Molecular Property Prediction

2026 – Present

Research Assistant | Advisor: [Dr. T. Ramraj](#)

- Developing efficient 3D graph kernel frameworks incorporating bond lengths, angles, and torsional features for molecular similarity modeling; investigating scalable 2D-to-3D kernel strategies to reduce time complexity.

Projects

Aurova: Mental Health Platform | RAG, LLMs, Full-Stack



- Privacy-first platform with anonymous journaling, AI counseling, and peer support and secure authentication.
- RAG-based LLM with crisis detection, safe response generation, and privacy-preserving context-aware therapeutic personalization.

LFW Face Generation & Latent Space Editing | AutoEncoders, VAE, Latent Interpolation, PyTorch



- Trained VAE models on LFW dataset for photorealistic face synthesis with latent space interpolation.
- Implemented attribute-driven latent vector editing to analyze disentanglement and geometry of the learned generative manifold.

HASYv2 Handwritten Math Symbol Classification | PyTorch, CNN, BatchNorm, Data Augmentation



- Built PyTorch CNN for 369-class symbol classification with custom preprocessing, BatchNorm, and adaptive pooling.
- Won 1st place in a university competition achieving 84% accuracy via hyperparameter tuning under constraints.

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript, Haskell

Full-Stack Development: Node.js, React, Next.js, Tailwind CSS, Express.js, MongoDB

Machine Learning & AI: PyTorch, TensorFlow, Deep Learning, CNNs, Transformers, NLP, Computer Vision

Libraries & Tools: NumPy, Pandas, Git, AWS

Certifications

AWS Cloud Practitioner Essentials – AWS (2025)

Deep Learning Specialization – DeepLearning.ai (2025)

Machine Learning Specialization – Stanford Online (2025)

Achievements & Leadership

Head Coordinator – Gokulashtami Event (2025)

Hackathon Winner – IETE Hackathon (2025)

AIR 270 – AEEE Exam (2023)

1st Place – Western Group Music, Gokulashtami (2025)